Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology

(B.Sc.I.T Semester IV)

MERN Stack

Practical – I

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Class: TYIT	Batch: 01
Date of Assignment: 20/06/2026	Date/Time of Submission: 20/06/2026

1.Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node -v
v24.17.0
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> npm -v
11.13.0
```

b. Create a simple Node.js file and run it using the terminal.

CODE:-

```
console.log("Welcome to MERN Stack");
console.log("Vinayak padhy ");
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node prac1b.js
Welcome to MERN Stack
Vinayak padhy
```

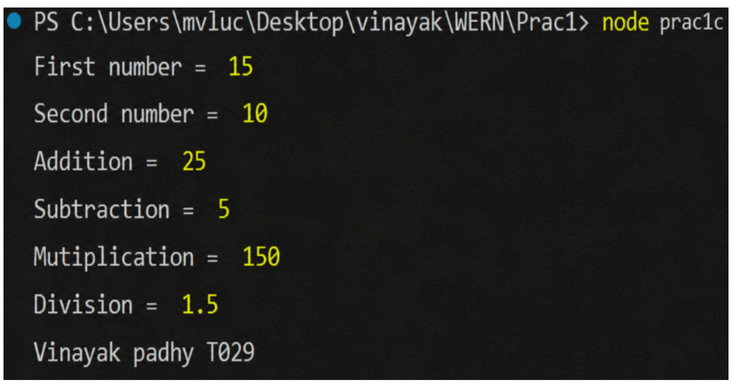
c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

CODE:-

```
let num = 15;
let num2 = 10;
console.log("First number = ", num);
console.log("Second number = ", num2);


console.log("Addition = ", num+num2);
console.log("Subtraction = ", num-num2);
console.log("Mutiplication = ", num*num2);
console.log("Division = ", num/num2);
console.log("Vinayak padhy T029")
```

OUTPUT:-


```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node prac1c
First number =  15
Second number =  10
Addition =  25
Subtraction =  5
Mutiplication =  150
Division =  1.5
Vinayak padhy T029
```

d. Create and run a simple program using variables and functions. **CODE:-**

```
let num = 15;
let num2 = 10;

function Add(a,b)
{
 return a+b;
}
function Sub(a,b)
{
 return a-b;
}
console.log("Addition = ",Add(num,num2));
```

```
console.log("Subtraction = ",Sub(num,num2));
console.log("Vinayak padhy T029")
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node prac1d.js
Addition =  25
Subtraction =  5
Vinayak padhy T029
```

e. Demonstrate use of operators and control statements in JavaScript **CODE:-**

```
let num = 15;
let num2 = 10;

if(num>num2)
{
 console.log("Number 1 is greater",num);
}
else{
 console.log("Number 2 is greater",num2)
}

let day = 2;

switch(day){
 case 1:
 console.log("Monday");
 break;
 case 2:
 console.log("Teusday");
 break;
 case 3:
 console.log("Wednesday");
 break;
 default:
 console.log("Invalid day")
}
console.log("Vinayak padhy T029")
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node prac1e.js
Number 1 is greater 15
Teusday
Vinayak padhy T029
```

f. Write programs using loops and template literals.

CODE:-

```
let num = 5;

console.log(`Multiplication Table of ${num}`);

for(let i = 1;i<=10;i++)
{
 console.log(`${num} x ${i} = ${num*i}`); }
console.log("Vinayak padhy T029")
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\vinayak\WERN\Prac1> node
prac1f.js
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
Vinayak padhy T029
```